

EXERCISE

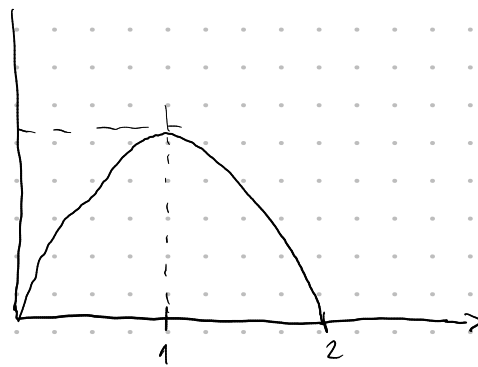
PROBABILITY DISTRIBUTION FUNCTION OF RV X

$$f(x) = \begin{cases} C(4x - 2x^2) & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

1) COMPUTE C

2) $P\{X > 1\}$

2 $\leadsto$ DRAWING WOULD ANSWER REQUEST EASY
THAT $P\{X > 1\} = 0.5$



1 $\leadsto$ C SHOULD SATISFY NORMALIZATION PROPERTY

$$\int_0^2 C(4x - 2x^2) = 1$$

$$2C \left[2x^2 - \frac{2}{3}x^3 \right]_0^2 = 1$$

$$2C \cdot \frac{4}{3} = C \frac{8}{3} = 1$$

$$\Rightarrow C = \frac{3}{8}$$

~~X~~: ~~HEADS~~ - ~~TAILS~~ MFLIP (FAIR COIN)

1) SUPPORT X

2) PMF X , $M=3$

SUPPORT
 $M=1$: $\{+1, -1\}$

$M=2$: $\{+2, 0, -2\}$

$$M=3: \{+3, +1, -1, -3\}$$

WE'D LIKE TO GENERALIZE

$$S = \{M-2j, 0 \leq j \leq M\}$$

PMF:

LET'S TRY WITH $M=3$

WE SHOULD FIND $P\{X=3\}$, $P\{X=1\}$

THEN IF THE COIN IS FAIR $P\{X=-k\} = P\{X=k\}$

$$P\{X=3\} = \frac{1}{8}$$

$$P\{X=1\} = P\{X=-1\} = \frac{3}{8}$$

$$\text{BUT ALSO } P\{X=3\} + P\{X=1\} = \frac{1}{2}$$

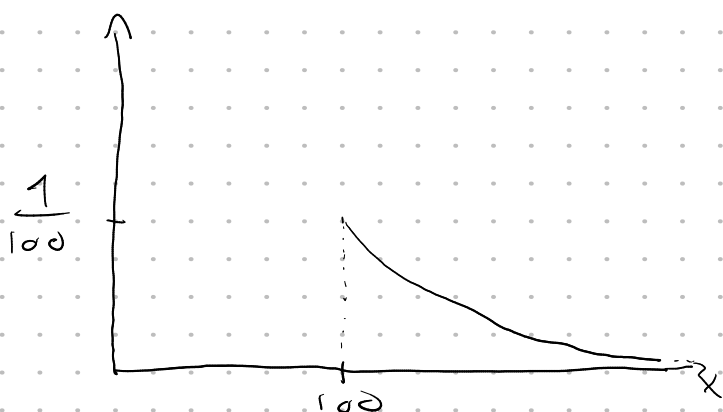
RADIO 5 TUBES

LIFETIME OF A TUBE:

$$f(x) = \begin{cases} 0 & x \leq 100 \\ \frac{100}{x^2} & x \geq 100 \end{cases}$$

$P\{\text{EXACTLY 2 TUBES/S REPLACED BY ISO}\}$

LIFETIMES OF TUBES ARE INDEPENDENT RVs



FIRST LET'S COMPUTE THE

$$P\{\text{ONLY IS REACHED BY ISO}\} = \int_{100}^{150} \frac{100}{x^2} dx$$

$$= -100 \left. \frac{1}{x} \right|_{100}^{150} = -100 \left[\frac{1}{150} - \frac{1}{100} \right] = \frac{1}{3} \quad \therefore p$$

NOW WE KNOW THAT $1-p$ IS THE PROB. THAT ONE DON'T RAIL

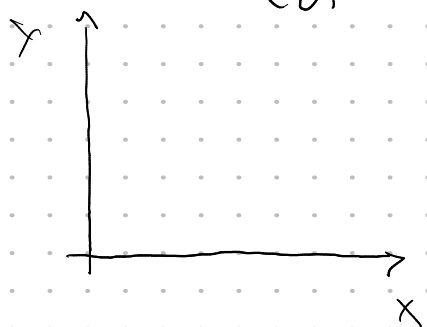
FFGGG $\rightarrow$ ONE POSSIBLE CONFIG.

$$p^2 (1-p)^3$$

ALL POSSIBLE CONFIGURATION ARE $\boxed{\binom{5}{2} p^2 (1-p)^3}$

JOINTLY DISTRIBUTED RVs

CDF



TO KNOW SOMETHING RELATED TO BOTH

YOU HAVE TO KNOW JOINT CDF

(IF THE TWO ARE NOT INDEPENDENT)

EXAMPLE: SMOKING AND CANCER: BOTH SUM ARE BUT THERE IS STRONG CORRELATION

$$\text{JOINT CDF} \quad F(x, y) = P\{X \leq x; Y \leq y\}$$

$$F_X(x) = P\{X \leq x; Y \leq +\infty\}$$

$$F_Y(y) = P\{X \leq +\infty; Y \leq y\}$$

FOR SRV WORKING NO

• **SPMF** (DISCRETE RM)

$$P(x, y) = P\{X=x, Y=y\}$$

$$P_X(x) = \sum_{y_s} P\{X=x, Y=y_s\}$$

SAME FOR $P_Y(y)$

DATA STRUCTURE FOR THIS COULD BE A MATRIX

$X \backslash Y$	y_1	y_2	...	y_k	
x_1	$P(x_1, y_1)$	$P(x_1, y_2)$	...		$\sum = P_X(x_1)$
x_2	$P(x_2, y_1)$	$P(x_2, y_2)$	...		$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$		$\vdots$
x_k	...	...	$P(x_k, y_k)$		$\sum = P_X(x_k)$
	$\sum_{x_i} P_X(x_i)$	$\sum_{x_i} P_X(x_i)$	...	$\sum_{x_i} P_X(x_i)$	

$P_X(y) \rightarrow P_X(y_1) \dots P_X(y_k)$

$P_X(x), P_Y(y)$ ALSO CALLED MARGINAL PROBABILITIES

CAN WE FIND SCDF FROM JPMF AND VICE VERSA?

YES!

$$F(x, y) = \sum_{x_1 \leq x} \sum_{y_2 \leq y} P(x_1, y_2)$$

• **JPMF**

X, Y

$f(x, y)$ IS JPMF IFF

$$\forall C = \{(x, y), \dots\}$$

$$P\{(x, y) \in C\} = \iint_{(x, y) \in C} f(x, y) dx dy$$

• **SCDF**

$$F(a, b) = P\{X \leq a, Y \leq b\}$$

$$\begin{aligned}
 &= P\{-\infty \leq X \leq a, -\infty \leq Y \leq b\} \\
 &= \int_{-\infty}^a \int_{-\infty}^b f(x, y) dy dx \\
 &= f(a, b) = \frac{\partial^2 f(a, b)}{\partial a \partial b}
 \end{aligned}$$

RECAP

$$P_{M.F} \leftarrow \sum JPMF \quad \text{so} \quad POF \leftarrow \int_{-\infty}^{\infty} JPMF$$

$$1) P\{X \in A\} = P\{X \in A, -\infty \leq Y \leq +\infty\} = \int_A \left[\int_{-\infty}^{+\infty} f(x, y) dy \right] dx$$

$$2) \hookrightarrow \int_A f_X(x) dx \quad f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy$$

EXERCISE

$$f(x, y) = \begin{cases} 2e^{-x}e^{-2y} & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

$$P\{X > 1; Y < 1\} \quad P\{X < Y\} \quad P\{X > a\}$$

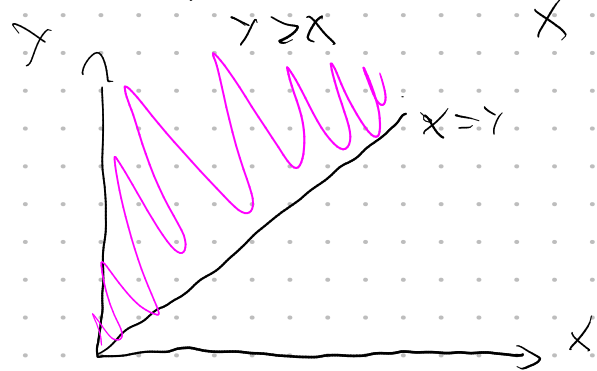
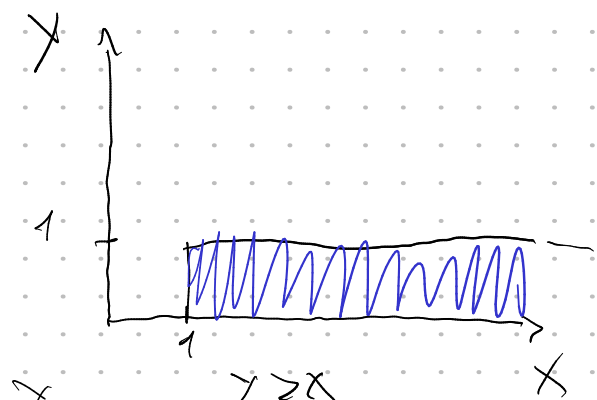
IT APPEARS TO BE HARD BRUISING THE BLOODY THING

$$P\{X > 1, Y < 1\} = \int_0^1 \int_1^{+\infty} 2e^{-x}e^{-2y} dx dy$$

$$= 2 \int_0^1 e^{-2y} \left[-e^{-x} \right]_1^{+\infty} dy$$

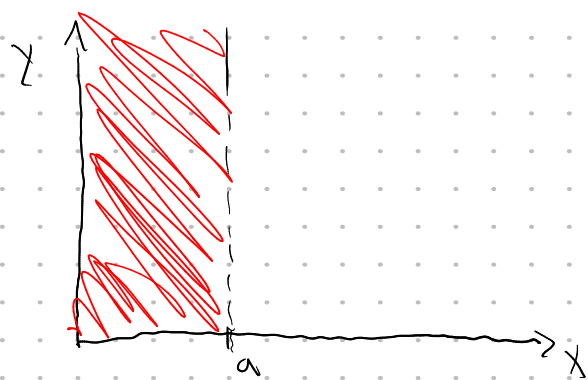
$$= \frac{2}{e} \int_0^1 e^{-2y} dy =$$

$$P\{X < Y\} = P\{X < Y, -\infty < Y < +\infty\}$$



$$= \int_0^{+\infty} \int_0^x (-2e^{-x} e^{-2y}) dy dx$$

$$P\{X \leq a\} = \int_0^{+\infty} \int_0^a f(x,y) dy dx$$



3 DISTINCT RV

$$X_1, \dots, X_m \quad F(x_1, \dots, x_m) = P\{X_1 \leq x_1, \dots, X_m \leq x_m\}$$

$$\swarrow$$

$$f_{x_i}(x) = F(\infty, \dots, x, \dots, \infty)$$

IF X, Y ARE INDEPENDENT THEN

$$F(x, y) = F_X(x) \cdot F_Y(y)$$

THIS BOA OF $X \leq x$ IS AN EVENT

$P\{X \leq x; Y \leq y\}$ IS AN INTERSECTION OF TWO EVENTS

$\downarrow$

$$P\{X \leq x\} \cdot P\{Y \leq y\}$$

$$= F_X(x) \cdot F_Y(y)$$

X, Y $\nwarrow$ PMF(?)

$\rightarrow$ DISCRETE $P(x, y) = P_X(x) \cdot P_Y(y)$

$\rightarrow$ CONT $f(x, y) = f_X(x) \cdot f_Y(y)$ $\nwarrow$ PDF(?)